Predicting Final-Hour SPX Price Movements for Short-Dated Options Trading Using Intraday Market Features

Harsh Khatri

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Contents

1 Overview

Often in the last hour of trading, financial markets frequently see increased volatility, especially in the S&P 500 index (SPX), where Zero-days-to-expiration (0DTE) options are extremely vulnerable to sudden price changes. Although it is challenging to consistently be able to predict these short-term movements, they can present significant trading opportunities.

This dissertation explores the possibility of using intraday market behaviour patterns to forecast notable price changes during the last trading hour. Machine Learning (ML) models are created to determine whether the market is likely to move significantly upward, downward, or remain relatively stable prior to market close by analysing historical price and volatility data.

Additionally, the study looks at whether these forecasts can enhance decision-making in a streamlined 0DTE options trading approach. The objective is to close the gap between statistical modelling and real-world trading applications.

1.1 Aim

To develop and evaluate ML models that predict the direction and magnitude of SPX price movements during the final trading hour, and to assess their usefulness in guiding short-dated options trading strategies.

1.2 Research Questions

This study aims to investigate the predictive power of intraday market features on end-of-day price movements in the SPX, as well as their applicability in options trading strategies. The research is guided by the following questions:

1. **RQ1:** Can intraday market features be used to predict the direction of SPX price movements during the final hour of trading?
2. **RQ2:** To what extent can intraday features predict the magnitude of SPX price movements during the final hour of trading?
3. **RQ3:** Which intraday features (e.g., momentum, volatility, and price positioning) are most important in predicting large end-of-day movements?
4. **RQ4:** Can SPX price movements in the final hour be predicted with sufficient accuracy to be economically meaningful for trading applications?
5. **RQ5:** Can these predictions be used to construct a profitable end-of-day options trading strategy after accounting for transaction costs and realistic execution constraints?

1.3 Motivation and Research Significance

Many traders and researchers believe that 0DTE options represent one of the best ways to speculate, hedge or mitigate potential risks associated with daily trading. Many recent studies on markets show that today 0DTE options are responsible for a majority of the SPX options that are traded; this may be the reason why it has become such an important factor for researchers studying day-to-day price behaviour TabbFORUM (2026), Cboe Global Markets (2026*b*).

Since 0DTE options' prices can fluctuate quickly in response to even small movements in the underlying index in the last hour of trading (due to rapid theta decay, high levels of gamma exposure or acceleration, and the fast moving delta), movements in the SPX over the last hour of trading (3:00 - 4:00 pm) can result in large percentage changes in the value of near-the-money and/or slightly out-of-the-money option prices. Since the final hour of trading offers a potentially greater risk/reward environment than other times throughout the trading day, examining short-term directional moves in price can provide researchers with new opportunities.

We designed the current study with the belief that SPX price movement at the end of the trading session may not be entirely randomly distributed. Research studies have shown that intraday return distributions vary throughout the trading session and may be influenced by such variables as volume. Kawaller et al. (1990) laid some of the early groundwork for understanding frequency relationships among volatilities of the SPX Index and SPX Futures. Later studies have expanded upon their findings, illustrating that ML models which use realised volatility and high-frequency financial data features can offer superior predictions for returns and volatility Zhang et al. (2024), Ferreira & Medeiros (2021).

Our central hypothesis is that features derived from intraday market behaviour contain sufficient information to improve prediction of future direction and magnitude of SPX price movements during the final hour compared to simple baseline models and that these results could support a more systematic approach to making decisions related to same-day 0DTE option trading.

2 Literature Review

This literature review discusses papers that relate to predicting final hour SPX price movements, and it will be organised around four major themes that assist in answering the research questions: intraday return and volatility predictability, high frequency financial forecasting using ML techniques, option-implied information and 0DTE market structure, and the gaps of previous studies. Instead of presenting each study individually, the literature review will address how the literature supports the design of this dissertation.

Table 1: Summary of key literature and relevance to the dissertation

Theme	**Study**	**Main contribution**	**Relevance to this dissertation**
Intraday volatility predictability	Andersen & Bollerslev (1998)	Demonstrates that volatility models can provide useful forecasts when evaluated appropriately, especially when using higher-frequency information.	Supports the assumption that intraday market behaviour contains structure that may be useful for forecasting SPX movement.
Intraday volatility predictability	Poon & Granger (2003)	Reviews volatility forecasting literature and shows that forecast performance depends on data frequency, model choice, forecast horizon and evaluation method.	Justifies evaluating the final-hour prediction problem using a horizon-specific framework rather than relying only on general daily volatility evidence.
Intraday market behaviour	Kawaller et al. (1990)	Documents intraday relationships between SPX futures volatility, SPX index volatility and trading volume.	Supports the use of OHLCV-derived predictors such as realised volatility, volume behaviour, intraday range and time-specific price movement.
ML and high-frequency forecasting	Ferreira & Medeiros (2021)	Shows that ML models can improve intraday return forecasting using minute-level market data, lagged returns, realised-volatility measures and CBOE Volatility Index (VIX) information.	Supports the use of non-linear supervised learning models and volatility-based predictors for short-horizon SPX forecasting.
ML and realised volatility	Zhang et al. (2024)	Compares HAR-type models with ML models such as penalised regression, XGBoost, MLP and LSTM for realised-volatility forecasting.	Supports testing multiple model types and accounting for intraday and time-of-day effects rather than relying on a single linear benchmark.
ML and market microstructure	Rahimikia & Poon (2026)	Finds that ML models using high-frequency limit-order-book and news-related variables can outperform traditional realised-volatility benchmarks.	Highlights the value of high-frequency features, while also showing a limitation of this dissertation because only OHLCV-derived features are available.
Option-implied information	Fan et al. (2026)	Shows that option-implied rough volatility information can improve realised-volatility forecasts beyond traditional HAR-RV/VIX-based models.	Supports the idea that options-market information contains useful predictive content, although this dissertation focuses mainly on observable intraday SPX features.
0DTE trading performance	Vilkov (2023)	Examines realised payoffs from 0DTE SPX option strategies and shows that returns can be skewed, path-dependent and sensitive to liquidity and timing.	Justifies evaluating predictions through an economic trading lens rather than relying only on classification or regression accuracy.
0DTE gamma risk	Dim et al. (2024)	Studies 0DTE trading, gamma exposure and volatility propagation, documenting links between market-maker positioning and subsequent realised volatility.	Supports the importance of short-horizon market conditions and motivates using volatility, momentum and range-based proxies when direct gamma data are unavailable.

Theme	Study	Main contribution	Relevance to this dissertation
0DTE market impact	Vasquez et al. (2025)	Documents relationships between lagged gamma-related positioning and subsequent realised volatility in the underlying market.	Provides further support for examining whether intraday market conditions before the close contain information about later SPX movement.
Jump risk and 0DTE options	Bozovic (2025)	Studies intraday jumps in SPXW 0DTE options and their implications for pricing and hedging same-day options.	Supports predicting both direction and magnitude because sudden final-hour moves can strongly affect 0DTE option outcomes.
Market relevance and practitioner evidence	Cboe Global Markets (2023, 2026*b*), Xu (2025), TabbFORUM (2026)	Documents the growth and market relevance of SPX 0DTE options.	Motivates the dissertation topic by showing that 0DTE options are now a major part of SPX options trading rather than a niche market.
Practitioner risk evidence	Numerix & Risk.net (2024)	Discusses practical risk issues linked to zero-day options, including intraday hedging pressure and short-dated gamma exposure.	Provides practitioner support for considering execution costs, hedging pressure and market frictions when evaluating a 0DTE trading strategy.

2.1 Intraday Market Behaviour and Volatility Predictability

A primary assumption of this dissertation is that intraday market behaviour includes identifiable structure. Without some degree of structure within short-horizon SPX movements, there is no logical reason for utilising ML to predict the final trading hour.Literature has demonstrated that while volatility persistence is a well established concept Andersen & Bollerslev (1998) but the extent to which volatility is predictable depends upon the chosen data, model, forecast horizon and/or evaluation methodology Poon & Granger (2003). Further, Andersen & Bollerslev (1998) provided empirical evidence showing that high-frequency data is capable of revealing volatility patterns that do not appear at lower frequencies.

Previous research has also shown that intraday volatility is associated with market activity and time-of-day effects in the context of SPX related markets. Kawaller et al. (1990) documented systematic relationships between futures volatility, index volatility and trading volume. While this study is not recent, its findings remain relevant as they establish that SPX intraday behaviour is not independent of trading activity. Thus, the study supports the inclusion of features such as realised volatility, volume behaviour, intraday range and time based price movement in the dissertation's model.

These studies together help us justify the methodology chosen for this dissertation, and they also support the notion that variables observed prior to 3:00 p.m. contain information about subsequent price behaviour. However, these studies have also illustrated that volatility forecasting ability does not necessarily imply either directional predictability or trading profitability. Thus, this dissertation builds upon the foundation established by these studies by examining whether intraday features can be used to predict both the direction and magnitude of SPX movement during the final hour.

2.2 ML and High-Frequency Financial Forecasting

The relationship between intraday market and final hour SPX movement is non-linear and complex. Therefore, it is unlikely that all of the potential features identified above (e.g. lagged returns, realised volatility, volume acceleration, Volume Weighted Average Price (VWAP) distance, intraday high-low positioning, etc.) will act independently. ML methods are particularly suitable for modelling non-linear relationships between features.

Ferreira & Medeiros (2021) found that ML models improved intraday return forecasting when compared to linear models when applied to minute-level market data they used lagged returns, realised-volatility measures and VIX information to find relative performance among different approaches (i.e. Random Forests vs. LSTM models). Their results provide support for employing both price-based and volatility-based predictors in this dissertation. Additionally, their study showed that VIX-related information can be beneficial for incorporating implied volatility. However, their study focused on minute-by-minute return forecasting as opposed to a clear final-hour SPX classification problem.

Zhang et al. (2024) compared traditional HAR-type (Heterogeneous Autoregressive models) models with ML models (penalized regression, XGBoost, multi-layer perceptron, and LSTM models) for realizing volatility forecasting. They also found that ML approaches generally outperformed traditional HAR-type models when considering intraday similarity. Similarly, Rahimikia & Poon (2026) reported that ML models using higher quality high frequency information (limit-order-book and news-related variables) were able to outperform traditional realised-volatility benchmarks over a substantial portion of the out-of-sample period. They also provided explainability results indicating that bid/ask/mid-price variables were important predictors.

Altogether, this literature supports the application of supervised ML models to intraday financial features. Also, above literature suggests that one should evaluate multiple model types rather than solely rely on a single linear benchmark. Notably, however, studies typically employ rich microstructure data (limit-order book, bid–ask spread, and news variables) whereas this dissertation employs only OHLCV-derived features and technical indicators due to constraints in accessing higher quality and often expensive data.

2.3 Option Implied Information and 0DTE Market Structure

Vilkov (2023) documented that when researching the payoffs from multiple 0DTE SPX options trading strategies, the returns generated by each strategy were extremely skewed towards negative returns, highly dependent upon the path taken during implementation, and highly dependent upon liquidity and timing of trade. In general, this demonstrates that regardless of the accuracy of statistical predictions, those predictions should be evaluated through an economic lens. For example, while a model might correctly predict direction, it would fail as a viable trading strategy if the size of the final hour move was too small; if the premiums paid for trading were too high; or if bid/ask spread and other transaction costs reduced profitability.

Dim et al. (2024) and Vasquez et al. (2025) published additional research documenting how important short term market conditions are to 0DTE market makers' performance when exploring 0DTE trading, gamma exposure (GEX), and volatility . The authors found statistically significant relationships between market maker gamma exposure and subsequent realised volatility as well as intraday market behaviour. Both papers report statistically significant associations between observed 0DTE activity and observed changes to the underlying market. Because obtaining direct gamma exposure data is often difficult, this dissertation uses OHLCV-derived proxies such as realised volatility, range expansion, momentum, and price positioning to represent state information regarding the underlying market.

Another key factor to consider for 0DTE options is ”jump-risk” due to potential loss from abrupt price movements near expiration, which can negatively affect same-day options traders. Bozovic (2025) investigated intraday jumps in SPXW 0DTE options and documented implications of discontinuous price movements on pricing and hedging same-day options. The results confirm that the final hour is an economically meaningful period to study since rapid large movements at the end of a session can greatly effect the success of same-day options.

In summary, there exists a substantial amount of literature suggesting that 0DTE options act differently than longer-term option contracts. Short-term options are heavily affected by several variables including: the proximity to expiration (i.e. time-to-expiration), GEX, jump-risk or gaps, and execution costs/trade-friction. The combined effects of all of these variables create a unique set of characteristics that determine the value of short-term options.

2.4 Market relevance and practitioner evidence

Although peer-reviewed research provides the bulk of academic reference materials to this topic, practitioner and trade reports help in establishing relevance of topic in current age. Cboe Global Markets (2023, 2026*b*), Xu (2025) reports clearly indicate the explosive growth of SPX 0DTE (same day expiration) trades and demonstrate that same day expiration options now constitute a significant portion of all SPX option volume. Also, Numerix & Risk.net (2024) have discussed the risks associated with zero-day options, such as intra-day hedging pressures and short-dated gamma exposure.

2.5 Limitations of existing research and research gap

While there is considerable evidence within the literature supporting each of the three areas addressed here, there are significant gaps remaining that I plan to answer in the dissertation. Firstly, while numerous intraday volatility studies have shown that high frequency market behaviour has some sort of structure, few have examined how to use this knowledge to accurately predict SPX price movement at a specific time. Secondly, the majority of ML studies evaluating predictive capabilities of various models utilising intraday financial data use statistical metrics such as mean absolute percentage error, directional accuracy etc. as a measure of performance; however, they rarely provide a link to a viable trading strategy. Finally, while option studies examining zero days to expiration (0DTE) typically examine the distribution of payoffs, none examine if OHLCV derived intraday features can actually predict the direction and magnitude of SPX price changes prior to closing.

The objective of this dissertation is to address each of these gaps by combining all three areas into one focused empirical study. This dissertation uses intraday OHLCV-derived features, technical indicators, and volatility measures to predict SPX price changes occurring within the last hour of trading and also checks them for a profitable strategy.

3 Data and Methods

This section showcases how market data was collected and prepared, the development of the predictive models and their translation into the outputs into the economic and options-based evaluations used for RQs. The main methodological priority throughout the project was to preserve the chronological order of the observations and ensure that information from after the 15:00 decision point could not influence the predictions.

3.1 Data Collection and Description

I used secondary intraday financial market data via the Massive API. I chose Massive over the other alternative, as my final project required a longer and more consistent history of one-minute observations. Additionally, Massive allowed me to collect this data programmatically, store it locally and audit each session individually.

I collected three underling market series: SPX, SPY, and VIX. In this case, SPX represented the market that I wanted to predict, and it was the final-hour return that I would use throughout all parts of the research. I decided to include SPY as part of my collection due to its high correlation with the S&P 500, along with its ability to provide me with real-time traded volume and VWAP-related information unavailable directly from the SPX index. Finally, I used VIX as a proxy for expected future volatility prior to the final hour of trading.

The primary dataset included all days from 18 July 2024 to 17 July 2026. An additional dataset starting from 15 Feb 2023 was also collected for use in the sensitivity analysis and exploratory LSTM experiment. To protect against accidentally overwriting the original dataset and outputs, I maintained both datasets independently. I identified the regular market hours as 09:30 to 15:59 ET. All predictive variables were generated using information available until the completion of the 14:59 bar; I designated 15:00 as the decision point and 15:59 as the end of the outcome period. After applying stricter modelling eligibility rules, I was left with 496 valid daily observations from the first 501 sessions that passed through the initial alignment stage.

Rather than requesting all of the data at once through a single large-scale download operation, I requested all of the data one business day at a time. Successful ticker-date requests were then added to their corresponding date-partitioned Parquet files and registered into DuckDB. A Log was used to document successful, empty and failed requests, allowing any interrupted downloads to begin where they previously left off without having to repeatedly retrieve already-completed dates. The API keys associated with these requests were stored separately from all project related files.

In order to complete RQ5, I retrieved minute-level bars for same-day SPX options after I had filtered down the candidate trading sessions based upon the results of the predictions made by the underlying model. The retrieval was limited to Calls and Puts for strikes closest to at the money and offsets ranging from five to thirty points out-of-the-money relative to the current SPX price level. There were a total of 15,279 option-minute records present in my dataset containing these observations. Although I requested twenty candidate sessions, there were only seventeen that ultimately provided contracts that could be used for creating entry and exit bars. These entries were trade derived OHLC and VWAP aggregates instead of historic bid/offer prices; therefore, I had to make explicit assumptions regarding costs of executions when conducting my backtests.

3.2 Data Cleaning and Pre-processing

The SPY, SPX and VIX series did not always contain observations at exactly the same timestamps. As such, I chose to use SPY as my reference clock based on the minute level since it included consistent observations from the regular session as well as volume in order to create the additional features needed later. Each row of SPY was then associated with the most recent (or oldest) possible observation of both SPX and VIX which existed at, or prior to the time stamp of, each corresponding SPY row. I applied a 2-minute tolerance when associating each row; if a later observation was chosen for a particular row, there would be no association for any later rows. Any matches outside of the tolerance would remain unassociated so as to prevent stale data from being propagated forward indefinitely. Overall, I successfully linked 194,495 minute-level rows across 501 total sessions.

Prior to creating the modelling table by date, I performed several validation tests to ensure that the datasets contained unique timestamps, no missing values, complete sessions and standard market days. A full regular trading session is defined as containing 390 one-minute observations. Sessions that are shorter than a full session (i.e., an early close) were eliminated from further analysis due to their windowed feature sets and last hour returns being incomparable with those found within a standard trading day. In addition to retaining sessions with a fully populated final two hours of SPX data, I also required that the final SPX observation exist at both 14:59 and 15:59, require no missing values during the final two hours of SPX data; have valid return(s) in the final hour. Lastly, prior to saving the modelling files, I validated that no duplicate sessions existed; that no non-finite values existed; and that all required target reconstruction steps had been taken.

To determine how dependent the findings were upon the completeness requirement of the VIX series, I developed two separate versions of the datasets (strict vs. relaxed). The strict dataset contained a complete set of VIX data for the final two hours of each session (n=434) whereas, the relaxed dataset contained up to 5 minutes worth of missing VIX data (n=486). However, even in the relaxed version of the dataset, the VIX series must contain valid observations at 13:59, 14:29, 14:44 and 14:59 with no older than two minutes. The relaxed version maintained the same requirements for the SPX series as did the strict version. Thus, I established a controlled tradeoff between data quality and sample size versus accepting every available session.

Lastly, I transformed the aligned minute-level data into one row representing each trading session. All predictor variables were constructed using information available no later than completion of the 14:59 bar.

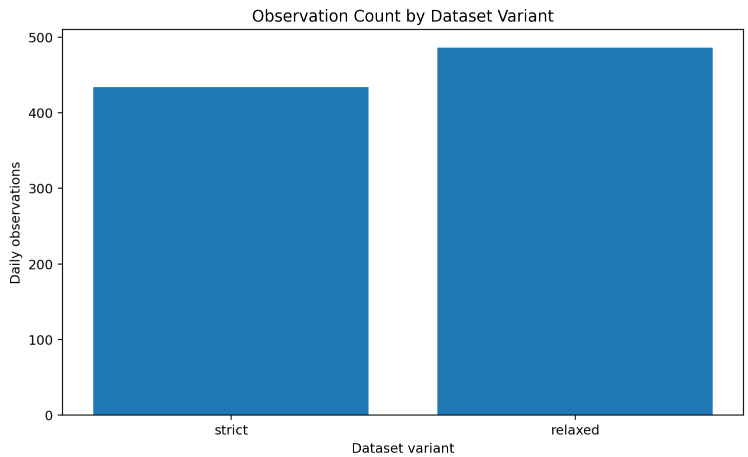

Figure 1: Number of daily observations retained under the strict and relaxed session-quality definitions. The relaxed rule increased the sample by permitting a small number of controlled VIX gaps while keeping the SPX requirements unchanged.

All of the features that included all of the data (22) included open-to-decision and recent returns, momentum acceleration, realised volatility, Average True Range (ATR), VIX levels and changes, SPY Volume and volume acceleration, the distance from VWAP, the position within the day's range of SPX and the distance from high-low of the session. I also created a smaller set of 15 features to be tested against each other with regards to reducing overlap so I could see if it would provide stability.

The last hour return is based on the % change from the 14:59 closing price of SPX to the 15:59 closing price. Zivot (2015)

$$r_d^{FH} = \frac{P_{d,15:59} - P_{d,14:59}}{P_{d,14:59}}, \tag{1}$$

In RQ1, a positive return was given a value of "1" for an up trade and a negative return as "0" for down. No sessions had a exactly a zero return in retained dataset.

$$y_d^{\mathrm{dir}} = \begin{cases} 1, & \text{if } r_d^{FH} > 0, \\ 0, & \text{if } r_d^{FH} < 0. \end{cases} \tag{2}$$

The absolute value of the last hour return was multiplied by 10,000 for RQ2 to measure the amount of movement in basis points.

$$y_d^{\mathrm{mag}} = 10{,}000 \left| r_d^{FH} \right| . \tag{3}$$

For RQ3 a large move was determined using the 75th percentile of the absolute value of the last hour return in the strict training sample.

$$y_d^{\mathrm{large}} = \begin{cases} 1, & \text{if } y_d^{\mathrm{mag}} \geq 27.5 \text{ bps}, \\ 0, & \text{if } y_d^{\mathrm{mag}} < 27.5 \text{ bps}. \end{cases} \tag{4}$$

As such, the approximate threshold of about 27.5 basis points was established in the training sample without utilising any information from either the validation or test periods.

Table 2: Chronological division of the strict and relaxed modelling datasets.

Split	**Date range**	**Strict**	**Relaxed**
Train	18 July 2024–3 December 2025	303	334
Validation	4 December 2025–20 March 2026	65	71
Test	23 March 2026–17 July 2026	66	81

The data were broken down into three parts: Training, Validation and Test (chronologically). A random split may have meant that a model trained on late market activity was being evaluated based upon early dates. Using the dates in time order will help avoid some types of "look ahead" bias.

Median imputation was applied to missing continuous feature values as the imputed value since it is less influenced by extreme or very unusual levels of observation compared to the mean. All continuous feature values below the 1st percentile of the training period or above the 99th percentile were clipped to those percentiles. Extreme values are considered to potentially represent one-time occurrences such as a single abnormal print or an error. The potential adverse effects on scaling and model development caused by extreme values can be mitigated through clipping. Deleting an entire trading session is not desirable, as it destroys all of the other valuable information contained in the session. The clipping percentiles (or limits) were computed solely based on the training data. After finding the 1st and 99th percentiles of the training set, these clipping limits were then directly applied to the related validation sets. No adjustments were made to the clipping limits based upon the validation set observations. By doing so, it ensured that no information from the validation data influenced how the data preparation occurred prior to model evaluation.

Scaling input to a model was also done in an identical manner as clipping. The scale parameters are computed independently with each train fold and then are applied to their respective validation folds. Both of the stored Strict and Relaxed datasets hold the original feature values instead of holding one permanent transformation of the data. Therefore, imputation, clipping and scaling can be performed at each validation stage utilising only the training observations available at that particular time.

Ultimately, final model comparisons were made primarily from evaluation metrics on the outer chronologically ordered folds. In each case, a model's performance was evaluated on dates not used when developing/fitting that particular model.

3.3 Modelling and Evaluation Method

The modelling process was organised around RQ1, RQ2 and RQ3, with a separate target and evaluation method used for each question. Before tuning more complex models, I first compared them with simple statistical and trading rules. This established a minimum level of performance that a machine-learning model needed to exceed. If a complicated model could not improve on a rule that required almost no fitting, its additional complexity would be difficult to justify.

For RQ1, the training-majority rule predicted the most common direction found in the training sample for every later session. The previous-session persistence rule assumed that the final-hour direction would be the same as it had been on the previous trading day. The momentum rules predicted that the direction observed over the 15 or 60 minutes before the 15:00 decision point would continue during the final hour. The mean-reversion rules made the opposite assumption: if the SPX had risen over the previous 15 or 60 minutes, they predicted a downward final hour, and vice versa. These rules were included because they represented simple explanations for short-term market behaviour and could be calculated using information available before the target period began.

RQ2 was compared with a constant prediction based on the median movement size in the training data. This showed whether the regression models reduced error beyond predicting a typical final-hour movement on every session. For RQ3, average precision was compared with the proportion of large-move sessions in the training sample. This prevalence provided the expected reference level when large movements were less common than ordinary sessions.

The baseline stage then compared several types of machine-learning model. Linear and regularised models provided relatively simple and interpretable starting points. Nearest-neighbour methods tested whether similar intraday conditions led to similar final-hour outcomes. Tree-based ensembles were included because they can identify non-linear relationships and interactions between momentum, volatility, volume and price-position features. Kernel models were also tested because they can create more flexible decision boundaries without requiring a deep-learning architecture.

Each candidate was evaluated using the strict and relaxed dataset variants and, where appropriate, the full and reduced feature sets. The strict dataset gave greater priority to complete and consistently aligned trading sessions, while the relaxed dataset retained more observations. Comparing both variants allowed me to examine whether stronger data-completeness rules were more useful than the additional sample size. All preprocessing remained inside the model pipeline so that imputation, clipping and scaling were learned again from the training part of each fold.

The observations represented sequentially ordered trading sessions, so expanding-window validation was used instead of random cross-validation. The first model was fitted using the earliest training period and evaluated on the next block of dates. The training window was then expanded before the model was evaluated on a later block. This better represented the way a model would be trained using historical information before being applied to future sessions.

Shortlisted candidates were assessed using nested expanding-window validation with three outer folds and three inner folds. Within each outer training period, the inner folds compared 30 randomly selected hyperparameter combinations. The strongest configuration from the inner search was then fitted using the available outer training data and evaluated on the following outer validation block. The outer-fold observations were therefore not involved in choosing the parameters used to predict them. Averaging performance across the three outer folds also reduced the chance of selecting a model because it happened to work unusually well during one short market period.

The primary evaluation measure differed between the three prediction problems. RQ1 used balanced accuracy because it calculated performance across both the Up and Down classes without allowing the more common class to dominate the result. Balanced accuracy was calculated as the arithmetic mean of sensitivity and specificity as defined by Brodersen et al. (2010)

$$\text{Balanced Accuracy} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right), \tag{5}$$

Where TP, TN, FP and FN denote true positives, true negatives, false positives and false negatives, respectively. This metric gives equal importance to the Up and Down classes. Precision, recall and F1-score were retained to show the types of directional errors made by each model.

RQ2 used mean absolute error, reported in basis points, because it represented the typical difference between the predicted and realised movement size in an interpretable unit. MAE and RMSE were calculated using standard forecast-error definitions from Hyndman & Koehler (2006).R^2 followed the approach of Campbell & Thompson (2008).

$$\text{MAE} = \frac{1}{N}\sum_{d=1}^{N} |y_d - \hat{y}_d|, \tag{6}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{d=1}^{N} (y_d - \hat{y}_d)^2}, \tag{7}$$

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{d=1}^{N} (y_d - \hat{y}_d)^2}{\sum_{d=1}^{N} (y_d - b)^2}. \tag{8}$$

Here, y_d is the actual absolute final-hour SPX movement for session d, $\hat{y}_d$ is the movement predicted by the model, b is the movement predicted by the training-median benchmark, and N is the number of evaluated sessions. All movements were measured in basis points.

RMSE was included to place more emphasis on larger errors, while Spearman correlation examined whether the model ranked quieter and more active sessions correctly. Out-of-sample R^2 was also reported to show how much variation was explained relative to a constant prediction.

RQ3 used average precision as its main measure because large movements occurred less frequently than ordinary sessions, it was calculated using the definition by scikit-learn developers (2026)

$$\mathrm{AP} = \sum_k (R_k - R_{k-1})\, P_k, \tag{9}$$

Where P_k and R_k are the precision and recall at threshold k. Average precision summarises the precision–recall curve and is appropriate when large-move sessions are less common than ordinary sessions. Unlike accuracy, average precision assessed whether genuine large-move sessions were concentrated near the top of the model's predicted probabilities. Precision, recall and F1-score were also recorded to show the balance between detecting more large movements and generating additional false signals.

Table 3 presents the final development configurations. The RQ1 and RQ3 probability thresholds were selected from chronological development-period predictions instead of being fixed automatically at 0.5.

The resulting thresholds were approximately 0.496 for RQ1 and 0.146 for RQ3.

Table 3: Final RQ1–RQ3 development configurations selected using nested chronological validation.

RQ	**Target**	**Data**	**Selected model**	**Primary metric**
RQ1	Up or Down direction	Relaxed, full	Histogram Gradient Boosting	Balanced accuracy
RQ2	Absolute movement in basis points	Relaxed, reduced	RBF Support Vector Regression	Mean absolute error
RQ3	Movement of at least 27.50 basis points	Strict, full	RBF Support Vector Classification	Average precision

After model selection, I performed fold-matched benchmark comparisons, permutation importance, feature-group analysis, regime checks and confidence-coverage analysis. Market-regime boundaries and confidence thresholds were calculated from development information only. These checks assessed whether the selected models behaved consistently enough to support the later economic analysis rather than relying only on their average scores.

A small LSTM was included as an exploratory extension using the separate 2023–2026 dataset. Each session was represented by a 120-minute sequence from 13:00 to 14:59 containing 16 price, volatility, volume, alignment and time features. The network used 32 LSTM units, a 16-unit dense layer, 20% dropout and chronological validation without shuffling. Scaling was fitted within each fold, and the latest development block was used to select the epoch count and classification thresholds. The LSTM was not treated as a replacement for the primary classical models because the number of daily sequences was limited and the refitted models had not been evaluated on a fresh holdout.

3.4 Economic and Options Evaluation

RQ4 examined whether the RQ1–RQ3 predictions identified market movements that were potentially meaningful for trading rather than merely producing acceptable statistical scores. I used the selected models' out-of-fold development predictions so that each session was assessed using a model that had not been fitted on that row. The analysis compared directional accuracy, realised movement size, signal coverage and the concentration of larger final-hour opportunities. Moving-block bootstrap intervals were used to reflect the chronological dependence between nearby trading sessions.

The combined rule carried into RQ5 required a session to fall within the most confident 30% of RQ1 predictions, have an RQ2 predicted movement of at least 20 basis points, and be identified as a large-move session by RQ3. RQ1 supplied the predicted direction: an Up prediction selected a Call and a Down prediction selected a Put. This rule was chosen during the development analysis and frozen before the option outcomes were examined. It was therefore treated as development evidence rather than an independently confirmed trading rule.

$$S_d = \begin{cases} 1, & \text{if all three selection conditions are satisfied,} \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

Here, S_d represents the combined trading signal for session d.

For each qualifying date, I identified same-day SPX Call and Put contracts at the nearest available at-the-money strike. Contracts positioned between 5 and 30 index points out of the money were collected for sensitivity analysis, but ATM remained the primary specification. Candidate dates and contracts were fixed before calculating any profit or loss.

Under the medium cost scenario created, the execution adjustment on each side was the greater of 0.10 option points or 5% of the relevant reference price. A commission of $1.50 per contract was applied at both entry and exit to simulate real life cost.

$$\begin{aligned} a(P) &= \max\left(0.10,\ 0.05P\right), \\ \Pi_i &= M_i\left[\max\left(0,\ P_i^{\text{exit}} - a(P_i^{\text{exit}})\right) - \left(P_i^{\text{entry}} + a(P_i^{\text{entry}})\right)\right] - 3.00. \end{aligned} \tag{11}$$

Here, Π_i is the net dollar profit or loss for trade i; P_i^{entry} and P_i^{exit} are the trade-derived reference prices; and $a(P)$ is the execution adjustment. M_i is the contract multiplier Cboe Global Markets (2026a), which was 100 for the SPX contracts used. The final \$3.00 represents commission of \$1.50 at entry and \$1.50 at exit.

The primary strategy purchased the contract using the first usable option-bar open between 15:00 and 15:05 ET and sold it using the final usable close between 15:55 and 15:59. Option values were converted into dollar amounts using the standard multiplier of 100.

Table 4 summarises the frozen primary backtest rules.

Table 4: Frozen signal, contract and execution rules used for the primary RQ5 backtest.

Component	**Primary rule**
Candidate session	Top 30% RQ1 confidence, RQ2 prediction of at least 20 basis points and an RQ3 large-move signal
Trade direction	RQ1 Up selected a Call; RQ1 Down selected a Put
Contract	Nearest available ATM 0DTE SPX option
Entry	First usable open between 15:00 and 15:05 ET
Exit	Final usable close between 15:55 and 15:59 ET
Contract multiplier	100
Execution assumption	Medium synthetic cost
Direction comparator	60-minute mean reversion
Primary strategy	Buy after 15:00 and hold to the final usable close

The strategy was compared with a 60-minute mean-reversion direction rule using the same dates, contracts and execution assumptions. Additional stop-loss, take-profit, time-based, break-even and trailing exits were examined only as robustness checks and were not used to replace the primary hold-to-close strategy. Further analysis examined profit concentration, maximum favourable and adverse movement, recovery after a stop was touched, random Call/Put directions and the timing of gains and losses during the hour. Multi-contract strategies were compared with holding the same number of contracts to close so that using more capital was not presented automatically as a better strategy.

3.5 Technology, Architecture and Reproducibility

The artefact was developed in Python using Jupyter notebooks in a local Windows environment. Pandas and NumPy were used for data manipulation, while Requests handled communication with the Massive API. PyArrow was used to write Parquet files, and DuckDB provided a lightweight analytical database for registering the partitioned files, this was done to ensure seamless integration between all notebooks. Matplotlib was used for EDA, scikit-learn was used for the classical models, and TensorFlow/Keras was used for the exploratory LSTM analysis.

The resulting end-to-end architecture is summarised in Figure 2. It separates data retrieval and processing from model development and the later economic evaluation, while retaining the intermediate datasets and evidence needed to reproduce each stage.

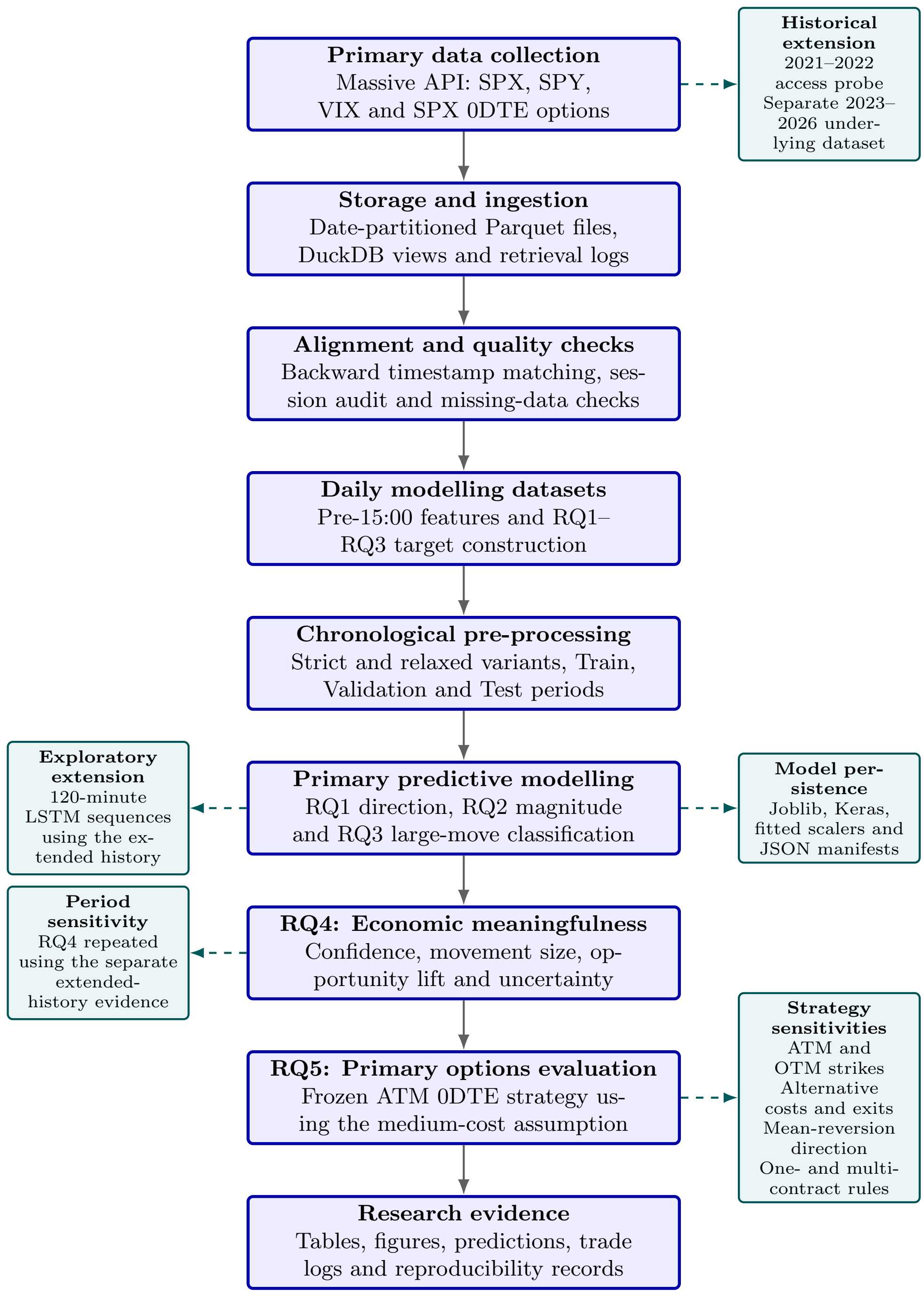

Figure 2: End-to-end architecture of the dissertation artefact. Solid arrows show the primary RQ1–RQ5 pipeline, while dashed branches show the extended-history, exploratory modelling, model-persistence and strategy-sensitivity work.

The project followed a staged architecture rather than placing the full analysis in one notebook. The retrieval layer collected and normalised API responses. The processing layer produced aligned minute data, session-quality audits and daily modelling datasets. Separate modelling notebooks handled baselines, tuning, robustness analysis, RQ4 and the RQ5 options backtest. All tables and figures were saved in the output directory, while fitted models were stored separately.

Repeated retrieval and storage operations were moved into a shared Python script so that the notebooks used the same pagination, retry, timestamp and database-registration logic. A central configuration file stored the project paths, market times, chronological split method and a random seed of 42. The API key was loaded from a separate environment file and excluded from version control.

The main notebooks were numbered in execution order, and each stage saved its intermediate data rather than depending only on variables held in memory. Final scikit-learn models were saved as Joblib files, while the LSTM models used the Keras format with separate fitted scalers. The saved classical models used Train and Validation only; the previously viewed Test period and data after 17 July 2026 were excluded from fitting.

Ethical Approval and Data Handling

An ethical approval application was completed in consultation with the project supervisor as part of the DkIT research ethics procedure. The study used commercially available secondary financial-market data and did not involve human participants, surveys, interviews or personal information. The completed application is included in the appendix. Research data were stored in the secured project environment, while API credentials were kept in a separate environment file and excluded from version control. The analysis was conducted for academic research purposes and should not be interpreted as personal investment advice or evidence of guaranteed future returns.

4 Results

This chapter provides the results obtained for five research questions. Initially, it presents the predictive efficiency of the models built for RQ1-RQ3, which includes final hour SPX direction, extent of movement and likelihood of great price movement. Then, it focuses on the significance of merging the results in RQ4, which tests if the identified results create a smaller grouping of profitable trading opportunities. In RQ5, it contemplates the opportunities generated by making use of SPX 0DTE options using realistic assumptions regarding execution costs. The last section presents a comparative analysis of the machine-learning methods versus alternative directional, exit and contract-selection rules. Finally, this chapter concludes by summing up the findings obtained throughout all research questions.

4.1 Predictive Results for RQ1–RQ3

The three predictive tasks were evaluated differently. Determining the direction in the last hour is still difficult, however determining how much the SPX will move or if there are any sessions worth investigating (large-move) was better determined through the use of these predictive models.

For RQ1, the frozen selected model for this question was a Selected Histogram Gradient Boosting classifier, which was used with the Relaxed Dataset and Full Feature Set. The Mean Outer-Fold Balanced Accuracy for this task was .544. The standard deviation for this task was roughly .07 over the chronological folds. The results from this task indicated that the model performed better than both the training majority, momentum and previous session rule; however, it failed to perform better than the 60 minute mean-reversion rule, which had a balanced accuracy of .569 on the same folds.

Therefore, although the RQ1 results indicate some level of directional structure, they do not demonstrate that the model's predictions of direction in the last hour were reliable. The model's performance varied across the chronological folds, demonstrating that the relationship was very time-sensitive.

For RQ2, the selected model was an RBF Support Vector Regressor. The selected model was trained using the relaxed reduced-feature dataset. The Mean Absolute Error for this task was 13.97 basis points. The training median benchmark had a Mean Absolute Error of 15.29 basis points. Therefore, the model resulted in an average error reduction of 1.32 basis points.

While the error is lower, the practical benefit is quite small. The model resulted in a Spearman Correlation of .373, indicating that it can separate sessions into those that have less movement and those that have more movement based upon their expected size of movement. However, the out-of-sample R-Squared is approximately .003. Therefore, while the model can effectively distinguish between sessions with quiet and busy days relative to each other, it does not accurately predict how many basis points the SPX will actually move.

RQ3 demonstrated the largest relative improvement among all three questions. The selected Strict Full-Feature RBF Support Vector Classifier was trained to determine whether the absolute move for the SPX during the final hour would exceed 27.50 basis points. Averaged Precision = 0.501 and Average Large-Move Prevalence = 0.275. Thus, the model seems to be able to place large move sessions closer to the top of its prediction rankings. However, the outer-fold Standard Deviation = 0.189 demonstrates that this outcome is not consistent throughout each evaluation period.

In addition, the feature analysis reflects the differences between each task. While momentum and SPX position within its daily range were significant contributors to RQ1, their directional contribution remains limited. Features such as True Range, SPY Volume, Realised Volatility, Price Position and VIX Behaviour were more effective features for RQ2 & RQ3. The association found here should be viewed as indicative of a predictive relationship rather than causal in nature.

In comparison to the classical models selected for each question, none of the exploratory LSTM models improved performance when evaluating on matching dates. In fact, for RQ3, the best-performing LSTM had an average precision of approximately .387 versus .535 for the corresponding classical model. Since the LSTMs performed worse than the classical models for all three questions, adding more complexity did not clearly offer advantages for this data set

Table 5 summarises the main out-of-fold results.

Table 5: Selected predictive model performance for RQ1–RQ3

Research question	**Selected model**	**Primary metric**	**Model result**	**Comparison benchmark**
RQ1: Direction	Histogram Gradient Boosting	Balanced accuracy	0.544 ± 0.070	60-minute mean reversion: 0.569 ± 0.035
RQ2: Magnitude	RBF SVR	MAE	13.97 ± 5.41 bps	Training median: 15.29 ± 7.75 bps
RQ3: Large movement	RBF SVC	Average precision	0.501 ± 0.189	Event prevalence: 0.275 ± 0.170

Note: Results are means and standard deviations across the chronological outer folds. Lower MAE is better for RQ2, while higher balanced accuracy and average precision are better for RQ1 and RQ3.

4.2 Economic Meaningfulness of the Combined Signals

To answer RQ4, I changed the focus of the analysis away from how well each individual model performed relative to a statistical benchmark. Instead, I evaluated if there were identifiable patterns in how well the combination of three frozen models would be able to provide more actionable trading signals for the analysts. In other words, even though one model might have outperformed a statistical benchmark, it may still not have provided a usable trading signal. Therefore, the goal of this portion of my study was to evaluate if the combination of the individual predictions would be able to eliminate some portion of the total sample by providing either more obvious directional information or larger realised price movement.

The use of the confidence filter did improve the quality of the directional information generated by the models, but it did not enhance the ability to predict which sessions would produce larger price movements. Specifically, among the common evaluation sample used in RQ1, those predictions that indicated the model was within the most confident 30% of RQ1 predictions achieved a balanced accuracy of .572, whereas the overall balanced accuracy across all sessions was .532. However, the average absolute price movement decreased from 22.80 to 20.46 basis points. Thus, it appears that having higher confidence regarding direction was not equivalent to predicting a stronger market opportunity.

RQ3 proved to be more useful as an opportunity filter. Those sessions that were identified as being more likely to experience a large price movement (i.e., average absolute returns = 36.38 basis points), were also much more likely to achieve a price move of at least 30 basis points (45.8%). Conversely, such an occurrence only happened 24.3% of the time across the entire sample. On the other hand, the directional balanced accuracy (i.e., correctly guessing the direction of a large price move) for sessions that were identified as likely to experience a large price move was very similar to chance (.508). Hence, although RQ3 provided guidance as to when larger price moves are more likely, it could not assist in determining which direction they will occur.

The most effective filter incorporated elements from all three research questions. Only those sessions where RQ1 provided directional predictions in the highest confidence category, representing the most confident 30% of RQ1 predictions, RQ2 predicted magnitudes of at least 20 basis points, and RQ3 identified these sessions as being likely to include large price movements were included in further analysis. As is evident from Table 6, only 20 sessions from the common set satisfied all three requirements. Thus, only 7.2% of the common evaluation sample included in this table were selected based upon all three criteria. Furthermore, these selected sessions experienced an average absolute price movement of 38.16 basis points. Finally, in 60.0% of the selected sessions, a price move of at least 30 basis points was observed; whereas such an event only occurred 24.3% of the time across the entire evaluation sample.

Table 6: Economic characteristics of the combined RQ1–RQ3 signals

Selection rule	**Sessions**	**Coverage**	**Balanced accuracy**	**Mean absolute move (bps)**	**Moves ≥ 30 bps**
All common sessions	276	100.0%	0.532	22.80	24.3%
RQ1 high-confidence predictions	85	30.8%	0.572	20.46	22.4%
RQ3 large-move prediction	107	38.8%	0.508	36.38	45.8%
RQ1 high confidence and RQ3 flag	33	12.0%	0.571	31.90	45.5%
Full frozen rule	20	7.2%	0.604	38.16	60.0%

Note: The full frozen rule required an RQ1 high-confidence prediction, an RQ2 predicted move of at least 20 basis points, and an RQ3 large-move prediction. Coverage is measured against the 276 sessions available across all three models.

The Moving-Block Bootstrap (MBB) reinforced the same overall trend as the MBB accounts for correlations between consecutive trading days. This analysis indicates the Combined Rule yields an estimated rate of large moves at about 63% or roughly 2.6 times the complete data set's estimate of large moves. However, the variability of both Directional and Return Estimates remained quite high since the rule selects a very limited number of trades. The Confidence Interval for Directional Capture also crosses Zero, implying it is impossible to conclude from this phase that there will be a positive Trading Return.

Therefore, RQ4 provides only partial insight; the combination of signals appears to provide a good means to reduce the database size by selecting a few sessions with relatively higher rates of Larger Moves. The results are stronger for Opportunity Selection than Direction Prediction, and the analysis does not account for Option Prices, Execution Timing or Transaction Costs. Therefore, the selected Sessions along with Frozen Decision Rules were forwarded to RQ5 where their actual trading performance can be assessed.

4.3 RQ5 Options Trading Results

RQ5 created an options trading rule from the previous research findings. Each session had to meet all three components of the composite filter established in RQ4 to be selected as a tradable day. First, the RQ1 forecast had to be very confident; Secondly, the predicted price movement from RQ2 had to be atleast 20 basis points; Thirdly, the session had to be classified as a "large move" from RQ3. RQ1 would determine if a call or put was to be bought. For the 20 sessions identified by this process, there existed usable option data for 17 transactions.

One at-the-money option contract was bought using the closest available opening price between 15:00 and 15:05 and sold using the final usable closing price between 15:55 and 15:59. Since the intention behind the design of this study was to test the utility of the predictive signals without including an additional stop-loss rule selected after viewing the option prices' paths, this simple entry/exit rule was maintained as the primary specification.

According to the assumptions regarding transaction costs, the main ML strategy produced a total profit of $9,551 over the course of 17 option trades. Among these trades, eleven were profitable with an overall win ratio of 64.7% and an average return on the cost of the premiums paid of 53.5%. The profit factor was 2.84 and the greatest dollar drawdown from the beginning to the lowest point experienced was -$2,143.20. Under more stringent execution fees, however, total profit decreased to $7,304.

Initially these results seem strong but are transformed by comparing them to the relatively simple directional benchmark. On precisely the same days, a simple directional 60-minute mean-reversion strategy produced $9,691.40 under moderate transaction cost assumptions and was subject to a lower dollar draw-down of -$1,604. This suggests that the RQ1 directional model added no incremental value over that benchmark, but it does not establish whether mean reversion could independently identify profitable trading dates.

Table 7 presents multiple alternatives that were studied in order to evaluate how changing either the contract or exit rule affected performance. All of these studies are considered sensitivity analyses rather than as new strategies optimized against the same 17 trades.

Table 7: Selected RQ5 strategy results under the medium-cost assumption

Strategy	Contracts	Win rate	Total P&L	Mean premium return	Maximum drawdown
Primary strategy and directional benchmark					
ML direction: ATM hold to close	1	64.7%	$9,551.00	53.5%	-$2,143.20
Mean reversion: ATM hold to close	1	64.7%	$9,691.40	55.9%	-$1,604.00
One-contract exit alternatives					
ML: −50% stop only	1	47.1%	$8,529.28	48.3%	-$2,007.84
ML: −50% stop and +100% target	1	52.9%	$3,025.71	17.7%	-$1,546.14
ML: +100% target only	1	70.6%	$5,173.40	29.8%	-$1,900.20
ML: fixed exit at 15:30	1	64.7%	$1,479.90	8.1%	-$1,841.00
ML: fixed exit at 15:45	1	64.7%	$8,529.20	45.9%	-$1,850.20
Multi-contract alternatives					
ML: two contracts, target and break-even runner	2	52.9%	$11,729.91	33.3%	-$3,850.39
ML: two contracts, target and hold runner	2	52.9%	$11,554.99	33.0%	-$3,553.99
ML: two contracts, target and trailing runner	2	52.9%	$10,258.59	27.8%	-$3,452.10
Out-of-the-money contract alternatives					
ML: 10-point OTM hold to close	1	47.1%	$5,721.20	56.2%	-$1,858.00
ML: 20-point OTM hold to close	1	29.4%	$1,522.30	15.1%	-$1,742.25
ML: 30-point OTM hold to close	1	11.8%	-$18.65	-41.6%	-$1,292.75

Note: All strategies were evaluated using the same 17 trading opportunities and the medium-cost execution assumption. ATM denotes the nearest available at-the-money contract. The multi-contract return is measured against the total initial premium paid, so its higher dollar P&L should not be interpreted as a better return without considering the additional capital and drawdown. The alternative exit and strike rules are sensitivity checks rather than separately optimised strategies.

The fixed stop-loss and take-profit strategy reduced the main result from \$9,551 to \$3,025.71. Although its maximum drawdown improved by approximately \$597, the rule closed several trades before their option premiums recovered. Among the trades that touched a loss of 50%, 44.4% later returned to at least their original entry premium and 22.2% subsequently reached a gain of 100%. This explains why the apparently sensible stop-loss rule performed worse than holding the contract to the planned close.

The fixed-time exits produced a similar result. Closing at 15:30 reduced total profit to \$1,479.90, while waiting until 15:45 retained \$8,529.20. This suggests that much of the value of the selected opportunities appeared later in the hour. The trailing-stop specification reduced drawdown but retained only \$1,075.93 of the historical profit.

The multi-contract rules produced higher dollar profits, but this came from purchasing more contracts rather than from a more efficient exit. The two-contract target-and-hold strategy made \$11,554.99, but simply purchasing two contracts and holding both to close would have produced \$19,102. The scale-out rule therefore gave up approximately \$7,547 of historical profit. It also required median starting capital of approximately \$1,893, compared with \$946.50 for the main one-contract strategy. The three-contract strategy increased total profit further, but its mean return on the total premium fell to 23.0% and its drawdown increased to \$4,561.48.

Moving farther out of the money reduced the entry cost but made substantial losses more common. At 30 points out of the money, the median capital requirement fell to approximately \$126.50. However, the strategy lost \$18.65 under the medium-cost assumption, only two of the 17 trades were profitable, and 82.4% of the contracts experienced a near-total loss. The cheaper contract was therefore more accessible to a small account, but it was not the more reliable strategy.

The main result was also concentrated in a small number of unusually successful trades. The single best trade contributed approximately 41.6% of the total profit. Removing the three best trades reduced the remaining profit to only \$425.90, while removing the five best trades changed the result to a loss of \$2,143.45. Bootstrap uncertainty was similarly wide, with the interval for mean trade profit crossing zero.

RQ5 showcased promising profitability over the historical dataset, but not enough to claim that the ML stratergy is profitable. The filters obtained from RQ1–3 were useful for identifying the correct session with valuable option movement.

4.4 Summary of Results

When results were taken together, it shows that the models were more useful for identifying sessions with the potential for a larger final-hour move than for consistently predicting market direction. Table 8 summarises the main finding for each research question.

Table 8: Summary of findings across the five research questions

RQ	Focus	Main result	Overall finding
RQ1	Final-hour SPX direction	The selected Histogram Gradient Boosting model achieved a mean balanced accuracy of 0.544. It showed some directional information but did not outperform the stronger 60-minute mean-reversion benchmark.	Weak and unstable directional evidence.
RQ2	Absolute size of the final-hour move	The selected RBF SVR reduced MAE from the benchmark value of 15.29 to 13.97 basis points. Its positive rank correlation suggested that it was more useful for ordering sessions than predicting the exact move.	Limited magnitude-ranking ability.
RQ3	Large-move prediction and its most useful features	The selected RBF SVC achieved average precision of 0.501 against event prevalence of 0.275. Intraday range, SPY volume, realised volatility, price position and VIX measures were among the more useful feature groups.	Useful as an opportunity filter, although feature importance varied
RQ4	Economic meaning of the combined signals	The complete frozen rule selected 20 of the 276 common sessions. Of these selected sessions, 60.0% produced a move of at least 30 basis points, compared with 24.3% across the full sample.	Promising opportunity selection, but not evidence of profit by itself.
RQ5	Performance when applied to SPX 0DTE options	The main one-contract strategy generated $9,551 under the medium-cost assumption. However, it was based on only 17 trades and performed slightly worse than the mean-reversion direction benchmark.	Historically profitable but inconclusive.

The data shows that intraday information regarding SPX, SPY & VIX prices may enable narrowing of the overall market to those sessions with increased activity during the final hour. The limitations of using the above-mentioned method to filter relationships to previously conducted studies and conclusion for the current study will follow.

5 Limitations

Many of the limitations for this research come from practical decisions made to allow completion of the project. For example, only commercially available minute-by-minute data was used in place of all the information about the order book. Other restrictions were due to the relatively short period of time represented by the data and the limited number of trading opportunities defined by the final rules. While these constraints will reduce the degree to which the results obtained may be generalised to other periods of time in financial markets or regarded as proof of a viable trading strategy, they do not eliminate their potential usefulness.

5.1 Constraints Relating to Data and Samples

One constraint is the length of the dataset as the data used for modelling included only the period from 18 July 2024 to 17 July 2026. Although there were many daily observations (over 400) as well as varying degrees of volatility throughout this period, it is still a very short period of time relative to what is considered typical in financial markets. There were no observations from a complete business cycle, nor from each type of market condition. Therefore, relationships established during this time may perform differently under long-term bear markets, crises in financial systems or extended periods of abnormally low volatility.

There were also data-quality issues when creating the modelling datasets. The strict dataset required a continuous record of VIX observations during the specified time frame, while the relaxed dataset would accept a limited amount of missing VIX observations. As a consequence, I was able to evaluate the trade-off between the quantity of data (i.e., sample size), and data quality, but neither version eliminates this issue entirely. For instance, the strict dataset excludes sessional data that may provide valuable SPX and/or SPY information even though it has no complete VIX observation. Conversely, the relaxed dataset allows for a few percent of uncertain VIX observations and, thus, provider gaps or stale observations may affect some of the created features.

Additionally, I used OHLCV-derived features primarily in this project. Since SPX is an index and does not report volume, I used SPY volume as a surrogate for trading activity in the marketplace. SPY is similar to SPX, but it does not perfectly capture trading pressures on the index or its options. The feature set included no full limit-order-book information, option-implied volatility surfaces, dealer gamma exposure, news sentiment or detailed macroeconomic announcement variables or darkpool data. All of these factors may include information relevant to significant end-of-day price movements but could not be incorporated into the data available for use in this study due to low availability.

When applying the models to options, the sample size became much smaller than when applying them to equities. The RQ1-RQ3 rule selected only 20 sessions from among the 276 common evaluation sessions. Additionally, usable option contracts existed for 17 of the above dates. Although this was beneficial for determining whether the filter could select a smaller group of possibly active sessions, only 17 trades exist and are insufficient for establishing stable profitability. Due to this limited sample size, a single unusual day in the marketplace can significantly impact total profit/loss amounts, win rates and maximum drawdowns.

5.2 Modelling and Validation Limitations

The expanding window validation method maintained chronological ordering in observations and minimised the possibility that models were trained on future information. However, there were only three outer-folds in the nested evaluation process. Consequently, model performance was estimated across only three major chronological blocks (evaluation periods) rather than across numerous different market regimes.

RQ1 indicated only weak evidence of directional predictability even though the model performance exceeded that of several simple benchmarks. The model's performance did not exceed that of the 60-minute mean-reversion benchmark indicating weak evidence of reliable directional forecasting capability.

Similarly, RQ2 indicated only modest improvement as the model exhibited some capability to rank sessions that were likely to exhibit small moves and those that were likely to exhibit greater market moves. But, out-of-sample (R^2) was approximately .003. So the magnitude model should be interpreted strictly as a tool for ranking sessions rather than as an accurate bases for point forecasts.

However, RQ3 demonstrated a better model performance. Average Precision was improved but its outer fold Standard Deviation shows that this result was not uniformly spread across all evaluation periods. For RQ3 a large move was determined using the 75th percentile of the absolute value of the last hour's return in the strict training sample. However, its chosen threshold of roughly 27.5 basis points may not continue to be appropriate if overall levels of market volatility change.

Classification thresholds for both RQ1 and RQ3 were determined from chronological predictions generated during development periods rather than from fixed 0.5 thresholds. Given class distributions exhibited in both studies, I believed this was a reasonable approach to create more suitable thresholds for my application-specific problems. Nevertheless, thresholds selected from one development period may not continue to be optimal in different market regimes. An additional longer forward evaluation would be necessary to assess stability in these values.

An additional limitation exists regarding Feature Importance Analysis (FIA). FIA assists in identifying input variables upon which RQ3's selected model relies most heavily, i.e., True Range, SPY Volume, Realised Volatility, Price Position and VIX measurements. However, demonstrating importance in a predictive model does not imply causality regarding why a later price movement occurred. Certain features also demonstrated greater stability across folds than others. The results indicate predictive associations within this dataset but do not necessarily imply causal relationships.

Final results for LSTMs should also be interpreted cautiously, as LSTM models generally require many more observations to estimate many more parameters than classical models. Daily datasets available for this study may have been too limited to enable LSTMs to learn the robust temporal relationships. So, Weak Performance does not suggest that LSTMs are incapable of addressing this problem but that the additional complexity added by LSTMs was not justified by the quality or quantity of data available for this study.

5.3 Trading and Execution Restrictions

Limitations related to historical NBBO bid–ask quotes were the main issues in the RQ5 backtesting. In addition to historical NBBO bid—ask quotes being unavailable, entry and exit reference prices were derived from minute level bars generated by trades rather than directly observable executable quotes. Estimates of the bid–ask spread, adverse executions, and commissions (Low, Medium and High) were made to try and account for the restrictions; however, there is no guarantee that actual fills will be better or worse than those estimated depending on liquidity, the width of the spread and how quickly the option price moves during the minute.

Minute level bars limit the detailed information contained in each time period. Minute level bars can display the open, high, low, close and volume; however, the exact order in which every price occurred or if there was enough liquidity available at that price cannot be determined. For example, since the premium for 0DTE options can fluctuate significantly during the last hour, the inability to reproduce slippage, partial fills, temporary spread widening, or the time delay between creating a signal and submitting an order makes the backtest less representative of real-world trading scenarios.

The traded profitability result in the study was only for 17 trades and it was extremely sensitive to a small number of unusually successful trades. Which again points out that this should not be treated as expected future profitability.

A similar problem exists when utilising the mean reversion comparison. Similar to the original RQ1-RQ3 filter criteria, it used the same set of dates and substituted only the original RQ1 Call or Put direction. It did not function as a different means of identifying which days to trade via mean reversion. So, although it appears that the ML direction provided no incremental value on the specified dates ($9,691.40), it does not indicate whether mean reversion identified the profitable trading opportunities independently.

Also, both the limited size and non-regularly scheduled nature of the trading sample limit justification for traditional annualised risk-adjusted metrics. Since any Sharpe ratio type or Sortino ratio type calculations performed on the returns of individual trades are comparative metrics rather than commonly used annualised portfolio metrics based upon a small sample size of trades and do not accurately reflect regularly timed daily returns. Therefore, they are incapable of reliably estimating long term risk adjusted performance.

Lastly, neither live nor forward paper trading was used to evaluate the strategy. As previously stated, the backtested strategy failed to replicate all of the practical issues associated with trading that retail traders encounter, including order reject errors, timing/latency problems encountered on trading platforms, changes in trading commissions, tax implications or difficulties experienced by traders like emotions in maintaining positions in rapidly moving 0DTE options. Therefore, the results suggest that the combination of signals may aid in identifying times when larger movements in options occur. However, they do not provide evidence that a completed/traded independent profitable trading system has been created. Longer periods without intervention (i.e., hold-out periods), use of accurate historic quote data and prospective paper trading would be necessary to provide stronger claims.

6 Conclusion

In summary, this research study was designed to find if pre-15:00 hours information can be used to forecast directionally the last hour's SPX price movement and if such forecasts are useful to develop an SPX 0DTE Options Strategy. Results were inconclusive. Models performed relatively well in forecasting sessions where there will likely be a greater price movement than in sessions that will see little to no price movement. But the models' ability to predict direction was inconsistent. The combination of the two sets of signals resulted in a selective opportunity filter. The small sample size of options and reliance upon a few very large winners means that this project cannot conclude that a viable, independently profitable trading system was identified due to the small dataset.

6.1 General Findings and Contributions

RQ1 dealt with forecasting the direction of the SPX price movement during the last hour of trading. The Histogram Gradient Boosting classifier that was chosen for final hour direction forecasting failed to perform better than the 60-minute mean reversion benchmark, which achieved 0.569. Additionally, the variance between the chronological folds demonstrated that the directional relationship was unstable throughout time. Therefore, RQ1 only found limited evidence that a ML based model could improve directional forecasting.

RQ2 involved the magnitude of the absolute final-hour price movement. The model demonstrated a positive Spearman correlation of 0.373 suggesting that it was able to somewhat effectively rank less active and more active sessions; however, the model's out-of-sample R-squared measure was approximately 0.003.

Thus, the model was much more effective at ranking sessions based on their expected activity level versus predicting the actual number of basis points.

RQ3 was focused on classifying whether the final-hour price movement exceeded approximately 27.5 basis points and what features most contributed to making this classification. Performance between folds was highly variable. Feature analysis indicated that true-range, SPY volume, realised-volatility, price-position and various VIX metrics were among the more valuable categories. Cumulative SPY volume exhibited the largest average individual importance, whereas ATR as a percentage of price represented the most consistent feature across the three folds. Collectively these results indicate that volatilities and market activities were significantly more effective in identifying sessions with higher probability of larger-than-average price movements than in forecasting direction.

RQ4 combined the three sets of signals together into a single fixed selection criterion. Of the 276 common evaluation sessions, only twenty met all three criteria. Sessions meeting all three criteria had an average absolute price move of 38.16 basis points, compared to 22.80 basis points for all sessions collectively. On sixty percent (60%) of sessions selected using the combined criteria, a price move of at least thirty basis points (30 bp) occurred; in contrast to twenty-four percent (24.3%) for all sessions considered. The combined selection criteria therefore appears to be useful as a mechanism for filtering for potential trading opportunities; however, RQ4 did not consider option prices or trading costs and therefore cannot assert that the strategy alone was profitable.

RQ5 applied each of the selected trading opportunities to SPX 0DTE options. Under the assumption of medium cost, the primary one-contract-at-the-money options strategy generated a historical profit of $9,551 over seventeen applicable trades. However, the 60-minutes mean-reverting strategy generated $9,691.40 on the same sessions selected for trading and produced smaller drawdowns than the ML direction model. Thus, this comparative result indicates that the filtered opportunities may have contributed more to the trading success through filtering for potentially successful trading opportunities than the ML Direction Model contributed through direction forecasting.

Moreover, the RQ5 comparative results were heavily skewed toward only a few successful trades. When the five best performing trades were removed from consideration, total profits turned negative and confidence intervals for mean profit per-trade crossed zero. Therefore, historical profits represent an encouraging finding for additional research as opposed to reliable proof of future performance.

6.2 Future Research Directions

There are several key areas that will need to be addressed in any follow-up research efforts to validate the findings of this project. First and foremost, it will be essential to conduct evaluations of the frozen models, threshold values and trading rules utilising longer periods of un-influenced time series data. During such evaluations, it will be critical that the model parameters remain constant to ensure that the new observations represent truly prospective testing.

It will also be beneficial to use historical NBBO or other sources of executable bid/ask prices in future studies. Such utilization would allow for a more realistic estimation of spreads, liquidity and slippage associated with executing transactions rather than relying upon synthetic estimates derived from assumed transaction costs.

A forward paper-trading approach could be used whereby signal generation, selected contract specification, available quote acquisition and trade execution outcomes are recorded without risking actual capital. Features representing implied volatility from option pricing models, dealer gamma exposure (from Gamma Scalping), Order Book Information and Scheduled Economic Announcements could be added to expand feature space.

Additionally, long-term data availability would enable examination of stability in feature importance rankings and optimal threshold levels across varying levels of volatility.

Separate assessments for Opportunity Selection and Directional Selection would also be recommended. For example, mean reversion could be assessed as a standalone strategy capable of selecting its own trading date(s).

Thus, future research efforts could assess if value creation comes solely from combining Opportunity Filtering and Directional Rules or if they are contributing factors.

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A Generative AI Usage Declaration

Generative Artificial Intelligence has been used to assist with the artefacts developed in this study by providing assistance throughout all stages of artefact development. These tools assisted the researcher in developing a workflow visualisation of their planned approach, designing a flow chart for planning how they would execute their plan, and assisting the researcher in debugging and streamlining the programming associated with their analytical pipeline. AI was also used to help with time management.

A.I. was NOT used as an alternative for my own analysis, interpretation of the findings, or make any independent academic judgements regarding the project. All project-related decisions made throughout the course of this study remained my responsibility. For example, what data to select, which features to engineer from that selected data, how to design the models that will be used to analyse the data, how to interpret the results of those analyses, and how to create the final document detailing the results of this study. ANY generated output produced using an A.I.-assisted tool was reviewed, edited and validated prior to its inclusion in either the artefact itself OR as supporting documentation for this study.

The primary purpose of utilising generative A.I. in this study was to improve upon the efficiency of various support-related processes. Examples include improving upon the overall organisation of the coding structure, identifying potential coding errors, creating initial drafts of workflow descriptions, and providing additional clarity on how the artefact development process occurred. NO private/confidential/personal/participant identifiable information was inputted into these A.I. tools. This study uses secondary publicly available financial data and does not contain human subjects.

A.1 Generative AI Prompt Log

The following prompts represent the types of generative AI support used during artefact development, coding troubleshooting, workflow planning, and project time management. AI-assisted support was used as a development aid only, and all outputs were reviewed, edited, and validated .

- How should I organise the notebooks so that data collection, cleaning, feature engineering, modelling and backtesting are completed in the correct order?
- Why might a DuckDB query return an empty result even though the associated Parquet files contain data?
- How can I calculate the final-hour SPX return using the 14:59 and 15:59 closing prices more effeciently?
- What checks should I use to identify duplicate timestamps in minute-level SPX, SPY and VIX data?

- How can I interpret the Massive API documentation when requesting minute-level aggregate bars for a specific ticker and date?
- Which classification models would be suitable for predicting whether the final-hour SPX movement will be Up or Down?
- How can I register multiple date-partitioned Parquet files as a DuckDB view?
- What could cause a notebook to use an outdated dataset after a newer version has been saved?
- How can I align SPX and VIX observations with the SPY timeline without matching a future observation to an earlier SPY row?
- Which outputs should be retained from each model so that its performance can be examined later?
- How should missing columns or unexpected column names be handled when the API response schema changes?
- How can I structure a reproducible data analytics artefact for predicting final-hour SPX movements?
- Why might an option contract contain an entry bar but no usable exit bar?
- How can I use a backward timestamp match with a two-minute tolerance when combining SPX, SPY and VIX data?
- What is the difference between MAE, RMSE, Spearman correlation and out-of-sample R^2 when evaluating a magnitude model?
- How can I check that every trading date is associated with only one row in the final daily modelling dataset?
- How should database connections be opened and closed across several Jupyter notebooks?
- How can I prevent a failed download from overwriting a previously saved and valid data partition?
- Which simple trading rules should be used as benchmarks for the RQ1 machine-learning models?
- How can API pagination be handled when the requested data cannot be returned in a single response?
- How should the project folders be divided between raw data, derived data, models, tables, figures and documentation?
- Why might two Parquet partitions store the same column using different data types?

- How can I compare a strict dataset with a relaxed dataset without changing the prediction target?
- What checks can confirm that all model features use only information available before the 15:00 decision point?
- How can SPY cumulative volume and recent volume acceleration be calculated from minute-level bars?
- Why is chronological validation more appropriate than a random train-test split for this dataset?
- How can a DuckDB table or view be refreshed after additional Parquet partitions have been downloaded?
- How should empty API responses be distinguished from failed requests, permission restrictions and dates with no available data?
- How can I convert the model outputs into Call, Put or no-trade decisions?
- What is a suitable way to record successful, failed and empty API requests so that downloads can be resumed?
- How can I identify sessions containing fewer than the expected 390 one-minute observations?
- Which regression models should be compared when predicting the absolute size of the final-hour SPX movement?
- How can I investigate a database error caused by a missing table, missing view or incorrect database path?
- How should the RQ1 confidence score be calculated from the distance between the model score and its selected classification threshold?
- How can I ensure that the API key is loaded from an environment file and not written into a notebook or uploaded to GitHub?
- How can strict and relaxed dataset observation counts be checked against the dates stored in the source data?
- What causes duplicate rows after merging model predictions by trading date?
- How can I calculate realised volatility over 30-minute, 60-minute and 120-minute intraday windows?
- How should model preprocessing be kept inside the pipeline so that imputation, clipping and scaling are fitted using training observations only?
- How can I use DuckDB to query the underlying data without loading every Parquet file into memory?

- How should options tickers be constructed from the underlying symbol, expiration date, option type and strike price?
- Why might a numerical column be read as text after several datasets have been combined?
- How can nested expanding-window validation be implemented with inner and outer chronological folds?
- How should the 75th-percentile large-move threshold be calculated using only the strict training sample?
- How can I check whether missing VIX values are caused by provider gaps, timestamp misalignment or a genuinely unavailable observation?
- Which feature groups should be considered when examining momentum, volatility, volume, VIX behaviour and price position?
- How can repeatedly used SQL queries and database-registration code be moved into shared Python functions?
- What should happen when an API request is interrupted after some, but not all, dates have been downloaded?
- How can the full 22-feature dataset be compared fairly with the reduced 15-feature dataset?
- How can percentile clipping be calculated from the training data and then applied unchanged to later observations?
- What checks should be performed before saving the final modelling dataset to confirm that the target variables are correct?
- How can I determine whether an options contract was genuinely unavailable or was missed because of an incorrect ticker format?
- Which metrics are most appropriate when large-move sessions are less common than ordinary sessions?
- How can the selected RQ1, RQ2 and RQ3 outer-fold predictions be joined using their common trading dates?
- Why might the number of rows returned by a database query differ from the number of rows stored in the original source files?
- How can relative paths be used so that the notebooks continue to work when the project folder is moved?
- How should a training-majority benchmark be created without using validation-period information?
- How can the most confident 30% of RQ1 predictions be identified from confidence-percentile ranks?

- How can the database be checked for duplicated ticker-date partitions?
- What should be included in a configuration file for market times, dataset paths, random seeds and model settings?
- How can median imputation be performed independently within each chronological training fold?
- How should average precision be interpreted when comparing the RQ3 model with the observed large-move prevalence?
- How can API requests be retried without causing the same data to be saved more than once?
- How can I calculate distance from VWAP and position within the daily SPX price range?
- Why might a model produce different predictions after it has been saved and reopened?
- How can model thresholds, feature names and training dates be stored alongside each saved model?
- How can I verify that the strict dataset requires complete VIX coverage while the relaxed dataset permits only the specified missing observations?
- Which model outputs should be used to choose the final RQ1, RQ2 and RQ3 configurations?
- How can RandomizedSearchCV be used without allowing the outer validation fold to influence hyperparameter selection?
- How should the performance of Histogram Gradient Boosting be compared with a 60-minute mean-reversion rule?
- How can incomplete or corrupted Parquet files be identified before they are registered in DuckDB?
- How can model probabilities be converted into classifications using a threshold selected from development-period predictions?
- What checks can confirm that the training, validation and test date ranges do not overlap?
- How can I create a smaller opportunity set by requiring the RQ1, RQ2 and RQ3 conditions to be satisfied together?
- How should the project handle a session in which SPX is available but SPY or VIX observations are incomplete?
- How can I determine whether a database join is one-to-one, one-to-many or many-to-many?

- What is the best way to save daily model predictions so that the selected sessions can be passed to the options backtest?
- How can the SPX level at 15:00 be used to identify the nearest available at-the-money option strike?
- Why might the model perform well in one chronological fold but poorly in another?
- How can feature importance be calculated without interpreting predictive importance as evidence of causality?
- How should permutation importance values be compared across chronological folds?
- How can the selected model be compared with a constant training-median prediction for RQ2?
- How can an output manifest record which datasets, models, tables and figures were produced by a notebook?
- What validation checks should be applied after combining the underlying SPX, SPY and VIX datasets?
- How can the first usable option-bar open between 15:00 and 15:05 be selected consistently?
- How can the final usable option-bar close between 15:55 and 15:59 be selected without changing the rule for individual trades?
- How should trade-derived option bars be described when historical bid–ask quotes are unavailable?
- How can low-, medium- and severe-cost execution assumptions be represented in the backtest?
- How should the adverse entry-price adjustment be calculated using the greater of a minimum point penalty and a percentage of the option premium?
- How can the adjusted exit price be prevented from becoming negative after execution costs are applied?
- How should the SPX contract multiplier and entry and exit commissions be included in net trade P&L?
- How can return on premium be calculated using the executed entry cost rather than the unadjusted reference price?
- How should win rate, profit factor and maximum drawdown be calculated from the trade-level P&L records?
- How can cumulative P&L be ordered correctly by trading date before maximum drawdown is calculated?

- How can at-the-money and out-of-the-money option contracts be compared without treating lower premiums as automatically better?
- How can stop-loss, take-profit and fixed-time exit rules be tested as sensitivity analyses rather than newly optimised strategies?
- How should a stop-loss and take-profit being reached within the same one-minute bar be handled when their exact order is unknown?
- How can I test whether contracts that touched a 50% loss later recovered to their entry premium?
- How can multi-contract strategies be compared with buying the same number of contracts and holding all of them to the close?
- How can profit concentration be measured when a small number of unusually successful trades account for much of the total P&L?
- How can leave-one-trade-out checks show whether the overall result depends on a single trading session?
- How can moving-block bootstrap intervals be used while retaining some of the chronological dependence between nearby sessions?
- How can a random-direction comparison help separate the value of opportunity selection from the value of directional prediction?
- Why should Sharpe-style and Sortino-style calculations from irregular individual trades not be presented as annualised portfolio statistics?
- How can existing notebook code be rewritten into shorter reusable functions without changing its calculations?
- How can additional validation messages be included when a required dataset, model or database file is missing?
- How should saved scikit-learn models be reopened and tested through their prediction interfaces?
- How can Keras models and their fitted scalers be saved as related but separate artefacts?
- How can SHA-256 hashes be used to record whether a saved model file has changed?
- Which tables best summarise dataset sizes, selected models, benchmark results and trading outcomes?
- How can figure dimensions, labels and font sizes be adjusted for inclusion in an Overleaf dissertation?
- How can tables and figures be saved automatically using stable and descriptive filenames?

- How should notebook outputs be checked against the numerical values reported in the dissertation?
- How can a Markdown file explain the purpose, required inputs, processing steps and outputs of each notebook?
- What information should be included in a Markdown guide describing the correct notebook execution order?
- How can Markdown documentation link to datasets, models, figures and tables using relative paths?
- How should model assumptions and limitations be documented in the corresponding Markdown files?
- How can long notebook outputs be summarised in Markdown without copying every printed table?
- How can a Markdown model-artifact guide record the selected features, thresholds, training dates and software requirements?
- How can a project README explain database setup and notebook execution to another researcher?
- How should database paths, API access requirements and unavailable private data be documented for reproducibility?
- How can the final output inventory be checked to confirm that all expected tables, figures, models and Markdown files are present?

B Github Project Link

Github link to the project including all code/reports:
`https://github.com/Ferreter/Intraday-Market-Signals-and-Final-Hour-S-P500-Predictability`

C Mahara Documentation Link

Mahara Documentation Link with meeting minutes and progress:
`https://mahara.dkit.ie/view/view.php?t=a3108f281f2b65a72a1b`

D Ethical Approval Documentation

This appendix contains the ethical approval documentation completed for the dissertation project. The project does not involve human participants, vulnerable groups, surveys, interviews, focus groups, or animal research. The research uses secondary financial market data and therefore does not collect personal or identifiable information.

DUNDALK INSTITUTE OF TECHNOLOGY
School of Informatics & Creative Arts
Ethical Approval Form for Research Projects

Researcher Name: Harsh Khatri Year: 2026 Course MSC in Data Analytics

Title of project : Predicting Final-Hour SPX Price Movements for Short-Dated Options Trading Using Intraday Market Features

Name of supervisor/s Zohaib Ijaz Date 25:03:26
(if applicable)

This application is to be completed by the researcher and where appropriate, in conjunction with the project supervisor. The lead researcher/supervisor is responsible for submitting the completed form to the appropriate Research Ethics Committee (details below)

Please note: If your submission is incomplete or unclear, your application will be returned to you, and your project may be delayed.

Section 1

Type of Researcher

Please tick the appropriate box below to indicate the type of researcher you are:

Undergraduate
(proceed to section 2) ☐

Completed Ethical Approval forms for undergraduate research should be submitted to the relevant Departmental Research Ethics Committee (DREC).

Drec.dcamm@dkit.ie – Department of Creative Arts Media and Music
Drec.dcsm@dkit.ie – Department of Computing Science and Mathematics
Drec.dvhcc@dkit.ie – Department of Visual and Human Centred Computing

Postgraduate
(proceed to section 3) ☒

Completed Ethical Approval forms for Postgraduate research should be submitted directly to the School Research Ethics Committee (SREC).

Srec.ica@dkit.ie

Staff
(proceed to section 3) ☐

Completed Ethical Approval forms for Staff research should be submitted directly to the School Research Ethics Committee (SREC).

Srec.ica@dkit.ie

ICA Research Ethics Approval Application Leathanach - Page 1

Section 2

Please complete questions 1-4 listed below.

	Human and / or Animal Research	YES	NO
1	**Does your research involve human participants other than the following[1]?** • Research using exclusively secondary sources. • Research using materials legally accessible to the public that have legal protection, e.g., record of court judgements, data archives. • Research using materials that are publicly accessible and where there is no reasonable expectation for privacy, e.g., books, published third party interviews. • Observations of human behaviour in public where (i) those being observed have no reasonable expectation of privacy, (ii) there is no intervention on the part of the researcher nor any interaction between the researcher and those observed, and (iii) individuals are not identifiable in the results. If '**YES**', please complete **B and C** below.		
2	**Does your project involve working with animals?** – If '**YES**', please complete **B and C** below. – Please note that for ethical consideration: 'Animals' are classed as vertebrate animals including cyclostomes and cephalopods (DIRECTIVE 2010/63/EU)		
3	**Does your project involve working with participants from any of the following categories?** • Minors (under 18 years of age) • People with learning or communication difficulties • Patients • People in custody • People engaged in illegal activities If '**YES**', please complete **D** below.		
4	**Does your project have any possible ethical implications other than those outlined in questions 1, 2 and 3?** If '**YES**', please complete **E** below.		

A. **I consider that this project has no significant ethical implications[2] to be brought through the ICA School Ethics Review Process**
Please complete Section 4

☐

[1] If there is any doubt, researchers should contact the Chair of the SREC.
[2] In determining significant implications, please consider all potential risks attached to this project. If there is any doubt, researchers should contact the Chair of the SREC.

B.

I. **Is this study part of a larger project that already has ethical clearance?**

YES / NO

If **YES** please answer question B II.
If **NO** please answer question C.

II. If this study is part of a larger project that already has ethical clearance, are you proposing any changes to the operational plan already ethically approved?

YES / NO

If **YES,** please complete *Sections 3 and 4.*
If **NO,** then please provide the project details below and complete *Section 4*.

Title of project with ethical clearance: ____________________________________

C. Could this project have ethical implications that should be brought before the appropriate ICA Departmental Ethics Review Committee as it will be carried out with human participants?

YES / NO

If **YES,** please complete *Sections 3 and 4.*
If **NO,** please complete *Section 4.*

D. I consider that this project may have ethical implications that should be brought before the School Research Ethics Committee as it will be carried out with human participants in a "vulnerable" category.
Please complete Sections 3 and 4 ☐

All research carried out with human participants in a vulnerable category must be referred by the Departmental Research Ethics Committee to the School Research Ethics Committee for approval.

E. Could this project have ethical implications, other than those previously outlined, that should be brought before the appropriate ICA Departmental Ethics Review Committee?

YES / NO

If **YES,** please complete *Sections 3 and 4.*
If **NO,** please complete *Section 4.*

Section 3

3.1 Application Form Checklist

Please complete Section 3 and provide additional information as attachments.

My application includes the following documentation:	**INCLUDED (mark as YES)**	**NOT APPLICABLE (mark as N/A)**
Recruitment advertisement		NA
Participant Information Leaflet		NA
Participant Informed Consent form		NA
Questionnaire/Survey		NA
Interview/Focus Group Questions		NA
Debriefing material		NA
Evidence of approval to gain access to off-site location		NA
Ethical approval from external organizations. If ethical approval from external organizations is pending give details below		NA
Details		

3.2 Project Details

a) **Lay description (Maximum 200 words)**

Please outline, in terms that any non-expert would understand, what your research project is about, including what participants will be required to do. Please explain any technical terms or discipline-specific phrases.

In the last hour of trading, financial markets frequently see increased volatility, especially in the S&P 500 index (SPX), where short-dated options are extremely vulnerable to sudden price swings. Although it is challenging to consistently predict these short-term movements, they can present trading opportunities. This dissertation explores the possibility of using intraday market behavior patterns to forecast notable price changes during the last trading hour. Machine learning models will be created to determine whether the market is likely to move significantly upward, downward, or remain relatively stable prior to market close by analyzing historical price and volatility data. Additionally, the study will assess whether these forecasts can enhance decision-making in a streamlined end-of-day options trading approach. The objective is to close the gap between statistical modeling and real-world trading applications.

b) Research objectives (Maximum 150 words)

Please summarise briefly the objectives of the research.

RQ1: Can intraday market features be used to predict the direction of SPX price movements during the final hour of trading? RQ2: To what extent can intraday features predict the magnitude of SPX price movements during the final hour of trading? RQ3: Which intraday features (e.g., momentum, volatility, and price positioning) are most important in predicting large end-of-day movements? RQ4: Can SPX price movements in the final hour be predicted with sufficient accuracy to be economically meaningful for trading applications? RQ5: Can these predictions be used to construct a profitable end-of-day options trading strategy, after accounting for transaction costs and realistic execution constraints?

c) Research location and duration

Location(s)/Population*	Dundalk Institute of Technology
Research start date	2nd March 2026
Research end date	September 2026
Approximate duration	6 months

* If location/Population other than DkIT campus/population, provide details of the approval to gain access to that location/population as an appendix.

3.3 Participants

		YES	NO	N/A
Do participants fall into any of the following special groups?	**Minors (under 18 years of age)**			**NA**
	People with learning or communication difficulties			**NA**
	Patients			**NA**
	People in custody			**NA**
	People engaged in illegal activities (e.g. drug-taking)			**NA**
Have you given due consideration to the need for satisfactory Garda clearance?				**NA**

3.4 Sample Details

Approximate number	NA
Where will participants be recruited from?	NA
Inclusion Criteria	NA
Exclusion Criteria	NA

Will participants be remunerated, and if so in what form? **NA**

Justification for proposed sample size and for selecting a specific gender, age, or any other group if this is done in your research. **NA**

3.5 Risk to Participants

a) Please describe any risks to participants that may arise due to the research. Such risks could include physical stress, emotional distress, perceived coercion e.g. lecturer interviewing own students. Detail the measures and considerations you have put in place to minimize these risks

b) What will you communicate to participants about any identified risks? Will any information be withheld from them about the research purpose or procedure? If so, please justify this decision.

3.6 Informed Consent

	YES	NO	N/A
Will you obtain active consent for participation?			NA
Will you describe the main experimental procedures to participants in advance?			NA
Will you inform the participants that their participation is voluntary and may be withdrawn at any point?			NA

If the research is observational, will you ask for their consent to being observed?			NA
With questionnaires, will you give participants the option of omitting questions they do not want to answer?			NA
Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?			NA
Will the data be anonymous?			NA
Will you debrief participants at the end of their participation?			NA
Will your project involve deliberately misleading participants in any way, or will information be withheld? **If you answer yes, give details and justification for doing this below.**			NA

a) Please outline your approach to ensuring the confidentiality of data (that is, that the data will only be accessible to agreed upon parties and the safeguarding mechanisms you will put in place to achieve this.) You should include details on how and where the data will be stored, and who will have access to it.

Publicly accessible data and saved on DKIT OneDrive with password and MFA with Private access to Supervision Team and Researcher

b) Please outline how long the data will be retained for, if it will be destroyed and how it will be destroyed.

1.Storage : DKIT one Drive

2. Access: Private

3. Communication : Teams and Outlook

4. Digital Platform Usage : Mahara, DKIT one drive

5. Data maintenance: None destroyed after project

Section 4

Researcher I have read and I understand the DkIT Ethics Policy available from:
https://www.dkit.ie/assets/uploads/documents/Research/Policies/DkIT%20Research%20Ethics%20Policy.pdf

Signed: _Harsh Khatri __ Print Name: Harsh Khatri Date: _25/03/26
(Researcher)

Supervisor: Applications for Ethical Approval of Undergraduate projects are forwarded to the Departmental Research Ethics Committee for approval or referral to the School Research Ethics Committee. Applications for Ethical Approval of Postgraduate and Staff projects are sent to the School Research Ethics Committee for Approval.

I have read and approved this form & information:

Signed: _________zohaib________________ Print Name: ______Dr. Zohaib Ijaz____ Date: _25/03/26
(Supervisor/Head of Department/ Research Centre Director/ Head of School)

There is an obligation on the researcher and/or supervisor to bring to the attention of the Departmental/School Research Ethics Committee(s): (a) Any issues with ethical implications not clearly covered by this form (b) Any ethical issues which may arise during the carrying out of the research; (c) Any ethically significant change made to the project after approval.

Section 5 (For office use only)

STATEMENT OF ETHICAL APPROVAL (FOR UNDERGRADUATE PROJECTS ONLY)

This project has been considered using agreed department procedures and is now:

Approved: ☐

Referred to the School Ethics Committee: ☐

Signed: ________________________ Print Name: ________________________ Date: ___________
(Chair of Departmental Research Ethics Committee/Head of Department)

STATEMENT OF ETHICAL APPROVAL

This project has been considered using agreed School procedures and is now:

Approved: ☐

Rejected (further information sought): ☐

Chair of School Research Ethics Committee
This project has been considered by the Ethics Committee and ethical approval is granted.

Signed: ________________________Print Name: ________________________ Date: ___________
Chair of School Research Ethics Committee